

25 Sep

Lecture

First Law of Thermodynamics for control mass

Law of conservation of energy

Control mass: A fixed mass system mass does not change

Control volume: A spatial region where mass can enter and exit

Heat: Energy transfer due to temperature difference

Work transfer: Energy transfer due to the force applied over a distance

Energy Equation for Control Mass

Instantaneous Rate form (small interval

$$E_{cv} = \dot{Q} - \dot{W}$$

$$\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W}$$

$\dot{E}$: Rate of change of total energy within the control mass.

Sign Convention

Heat ($\dot{Q}$) $\rightarrow$ when heat is entering the system
+ve

Work ($\dot{W}$) $\rightarrow$ when work is done by the system
+ve

Energy Equation for finite changes

Consider total energy changes over a process from state 1 $\xrightarrow{t_1}$ to state 2 t_2

Starting point

$$dE_{cv} = \delta Q - \delta W$$

$\xleftarrow{\text{change in energy}}$

Here we start from instantaneous state

$$\int_{t_1}^{t_2} dE_{cv} = \overset{\text{delta}}{\int_{t_1}^{t_2} \delta Q} - \int_{t_1}^{t_2} \delta W$$

$$E_2 - E_1 = Q_{1-2} - W_{1-2}$$

$\xleftarrow{\text{subscript (describe states)}}$

$\rightarrow$ Total heat added from
State 1 $\xrightarrow{\text{to}}$ state 2

Energy Equation (Detailed)

$$KE = \frac{1}{2}mv^2$$

$$PE = mgh$$

$$E_{\text{total}} = U + KE + PE$$

$$E_2 - E_1 = (U_2 - U_1) + \frac{1}{2}m(v_2^2 - v_1^2) + mg(h_2 - h_1)$$

Question A gas in piston cylinder absorbs 500 J of heat and 200 J of work during expansion. Calculate change in internal energy.

$$q = +500 \text{ J} \quad w = 200 \text{ J (system working)}$$

$$U = q - w$$

$$= 500 - 200$$

$$q = 300 \text{ J}$$

State function

Depend only on the state of the system e.g. U, E
As state U_2, U_1

Path function

Depend on the specific process path taken e.g. Q, w
As path follow Q_{1-2}

Pressure Volume Work

work is force required to perform a task in some distance

As Pressure = $\frac{F}{\text{Area}}$

Area

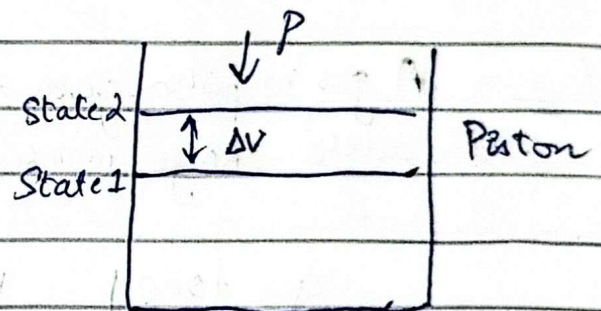
Volume = Area $\times$ distance

$\hookrightarrow m^3$

$\hookrightarrow \text{Length/distance}$

$$\begin{aligned} \delta W &= PdV \\ \delta W &= Fds \\ \delta W &= \frac{F}{dA} dA ds \\ &\quad \downarrow \quad \downarrow \quad \downarrow \\ &\quad \text{Specific Area} \quad \text{Area} \quad \text{Length} \\ &\quad \downarrow \quad \downarrow \quad \downarrow \\ &\quad m^2 \quad m \end{aligned}$$

$$\delta W = Fds$$



Isothermal (Temperature same)

(Pressure, Volume can change)

$$PV = nRT$$

Isothermal equation of an ideal gas (in terms of work)

Isothermal Process: Constant Temperature

Isobaric Process:

Constant Pressure

$$W = P(V_2 - V_1)$$

Adiabatic Process:

Lecture 10

First Law of Thermodynamics — Control Mass and cyclic process

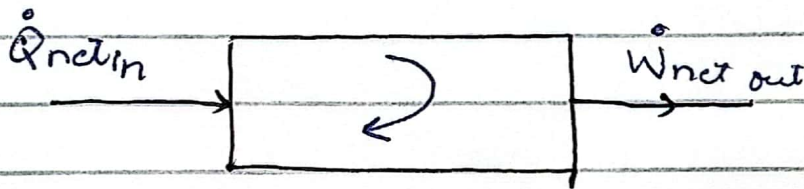
$$U = Q - W$$

$$0 = \oint Q - \oint W$$

It implies that there is no net change in the total energy of control mass and the rate of change of energy is zero

$$\boxed{\oint Q = \oint W}$$

Cyclic integral of heat transfer
Cyclic integral of work done



Problem: Engine absorb ³⁰⁰250 J heat from reactor and release 150 J of heat through radiator. Net work done by first law of thermodynamics and efficiency (cyclic process)

As cyclic process

$$\oint Q = \oint W$$

$$\oint W = 250 - 150$$

$$\oint W = 100 \text{ J}$$

$$= 300 - 150$$

$$= 150 \text{ J}$$

$$\text{Efficiency} = \frac{W_{\text{net}}}{Q_{\text{net in}}}$$

$$\frac{150}{300} \times 100 = 50\%$$

$$\text{Efficiency} = \frac{\text{output}}{\text{Input}} \times 100$$

$$= \frac{100}{250} \times 100 = 40\%$$

Problem

$$P_1 = 300 \text{ kPa}$$

$$V_1 = 0.05 \text{ m}^3$$

temperature same

The gas undergoes an isothermal expansion until its volume doubles. During expansion the gas absorbs 200 J of heat from surroundings.

a) $P_2 = ?$

b) Calculate the work done by the gas during expansion.

c) What is the internal energy change ΔU ?

a) $P_1 V_1 = P_2 V_2$

as $V_2 = 2V_1$

$$P_1 V_1 = P_2 2V_1$$

$$P_2 = 150 \text{ kPa}$$

b) $W = nRT \ln\left(\frac{V_2}{V_1}\right)$ If temperature and n is given

$$W = P_1 V_1 \ln\left(\frac{V_2}{V_1}\right)$$

$$= (300 \text{ kPa}) (0.05) \ln\left(\frac{2(0.05)}{0.05}\right)$$

$$W = 10.4 \text{ kPa}$$

c) Isothermal $\rightarrow$ temperature constant

So for ideal gas under isothermal process

$$\Delta U = 0$$

insulated material $\rightarrow$ no heat exchange

Insulated material $\rightarrow Q=0$

Adiabatic process

$$\Delta Q = Q - W$$

$$\boxed{\Delta U = -W}$$

$$\boxed{\Delta U = -10.4 \text{ kPa}} \quad \text{in insulated system}$$

If system is insulated then internal energy decreases

Problem A car has mass = 1100 kg KE 400 kJ find the velocity

$$KE = \frac{1}{2}mv^2$$

$$\sqrt{\frac{KE \times 2}{m}} = v$$

$$400 \text{ kJ} = \frac{1}{2}(1100)v^2$$

$$\sqrt{\frac{400 \times 10^3 \times 2}{1100}} = v$$

$$\boxed{v = 26.96}$$

How high would this system should be lifted have PE = KE

$$PE = mgh$$

$$PE = KE$$

$$PE = mgh$$

$$mgh$$

$$400 \text{ kJ} = mgh$$

$$(1100 \times 9.8)$$

$$\boxed{h = 37.105 \text{ m}}$$

Problem A stone having mass of 20kg and bucket contain 100kg of liquid water. The stone is at 10.2m above the water. The stone and water are at same temperature. (Stage 1) The stone then fall into water determine ΔU , ΔKE , ΔPE , Q and work done for following changes of states. State (2) The stone is about to enter the water. (State 3) stone has just come to rest in the bucket. (State 4) Heat has been transfer to surrounding in such amount that stone and water are at same temperature T_1 .

State 1 $\longrightarrow$ State 2

a) Temperature const (isothermal)

$$\Delta U = 0$$

b) $Q_{1-2} = 0$ (no heat exchange take place)

c) $W_{1-2} = 0$ (as $\Delta U = Q - W$)

$$\Delta U + \Delta KE + \Delta PE = Q - W$$

$$\Delta KE + \Delta PE = 0$$

$$\Delta KE = -\Delta PE = -mg(h_2 - h_1)$$

$$= -mg(h_2 - h_1)$$

$$= -(10)(9.8)(0 - 10.2) \text{ (potential energy decrease)}$$

$$\Delta KE = 1000 \text{ J}$$

$$\Delta PE = -1000 \text{ J}$$

State 2 $\longrightarrow$ State 3

$$Q = 0$$

$$W = 0$$

$$\Delta PE = 0$$

$$\Delta U + \Delta KE + \Delta PE = Q - W$$

$$\Delta U + \Delta KE = 0$$

$$\Delta U = -\Delta KE$$

$$\boxed{\Delta U = -1 \text{ kJ}}$$

State 3 — State 4

$$\Delta KE = 0$$

$$\Delta PE = 0$$

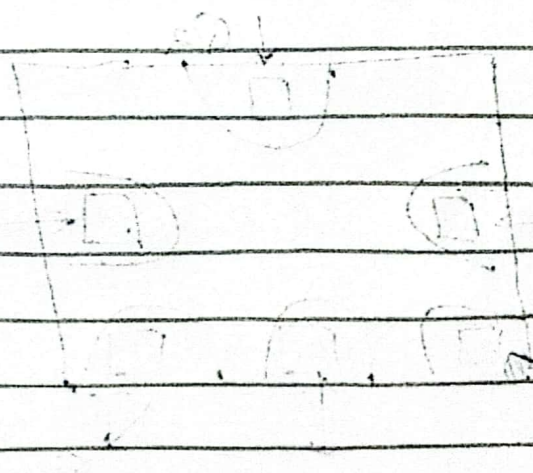
$$W = 0$$

$$\Delta U + \Delta KE + \Delta PE = Q - W$$

$$\Delta U = -1 \text{ kJ}$$

$$\Delta U = Q$$

$$\boxed{Q = -1 \text{ kJ}}$$



9 Oct 2024

Lecture 11

Modes of Heat Transfer

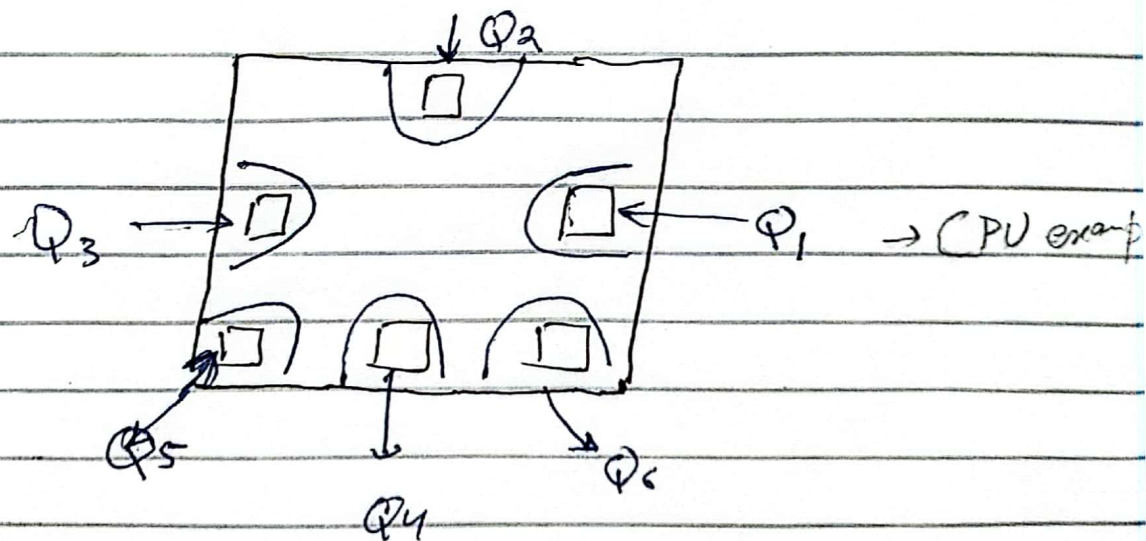
Energy transferred across the boundary of the system from one body to another (surroundings) due to temperature difference between two systems. Heat naturally flows from a system at high temperature to one at a lower temperature.

If heat does not flow $\rightarrow$ Thermal Equilibrium

Unit Joule (SI), BTU, calorie

Sign Convention

Heat into the system +ve
Heat out of the system -ve



Path function

$$Q_{\text{total}} = \int \delta Q$$

1) Conduction (Heat transfer due to direct contact of body)
Transfer of heat due to the direct contact between particles of matter. It happens because higher energy particles (those at higher temp) transfer energy to lower energy molecules (those at lower temp) via molecular collisions.

Fourier's Law of conduction:

$$\dot{Q} = -KA \frac{dT}{dx}$$

- $\dot{Q}$ = Rate of heat transfer
- K = Thermal conductivity of material
- A = Cross sectional area through heat is transferred
- $\frac{dT}{dx}$ = Temperature gradient

K (for metals) : ^{about} $100 \text{ W/m}\cdot\text{K}$

K (for Insulators) : upto $0.01 \text{ W/m}\cdot\text{K}$

2) Convection Heat transfer through a moving fluid (Liq or gas). It is combination of bulk motion of fluid and conduction.

2 types of convection

↳ Natural

↳ Forced

Newton's Law of cooling:

$$\dot{Q} = hA\Delta T$$

$h \rightarrow$ convective heat transfer coefficient

1) Natural Convection environmental changes

2) Forced Convection due to fan environment change

3 Radiation Heat transfer due to electromagnetic waves
and does not require any medium
e.g. Heat from sun that reaches Earth

Stefan - Boltzmann Law

$$Q = \epsilon \sigma A T_s^4$$

ϵ = emissivity of surface (0 to 1)

σ = Stefan - Boltzmann constant ($5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$)

A = Surface area

T_s = surface temperature

Problem conduction through Glass
Convection from glass to ambient air:

Room temperature = $T_{\text{room}} = 20^\circ\text{C}$

Outside temperature = $T_{\text{ambient}} = -10^\circ\text{C}$

outer glass surface temp: $T_{\text{outer}} = 12.1^\circ\text{C}$

Glass thickness: $x = 0.005 \text{ m}$

$$K = 1.4 \text{ W/m}\cdot\text{K}$$

$$A = 0.5 \text{ m}^2$$

$$L = 100 \text{ W/m}^2\cdot\text{K}$$

$$\dot{Q}_{\text{conduction}} = -KA \frac{\Delta T}{\Delta x}$$

$$\Delta T = 20 - 12.1 = 7.9^\circ\text{C}$$

$$\Delta x = 0.005 \text{ m}$$

$$\dot{Q}_{\text{conduction}} = - (1.4 \text{ W/m}\cdot\text{K}) (0.5 \text{ m}^2) \left(\frac{7.9}{0.005} \right)$$

$$\dot{Q}_{\text{conduction}} = -1106 \text{ W}$$

$$\rightarrow \text{rate of change of heat } \left(\frac{\text{J}}{\text{s}} \right)$$

$$\boxed{W = \frac{\text{J}}{\text{s}}}$$

$$\dot{Q}_{\text{convection}} = hA\Delta T$$

$$\Delta T = \text{outer glass} - \text{ambient air}$$

$$= 12.1 - (-10)$$

$$\Delta T = 22.1^\circ\text{C}$$

$$\dot{Q}_{\text{convection}} = hA\Delta T$$

$$= (100 \text{ W/m}^2\cdot\text{K}) (0.5) (22.1^\circ\text{C})$$

$$= 1105 \text{ W}$$

Extensive vs Intensive properties

Extensive Properties → depend on the amount of matter
(mass, volume, internal energy)

Intensive Properties → independent of the mass of the system (temperature, pressure)

Internal Energy

$$\begin{array}{ccc} U = m u & \longrightarrow & \text{specific} \\ \downarrow & \downarrow & \text{internal} \\ \text{total} & \text{mass} & \text{energy} \\ \text{internal} & & \\ \text{energy} & & \end{array}$$

$$u_{fg} = u_g - u_f$$

$$U_{fg} = U_g - U_f$$

Internal energy in the liquid vapour region

$$U = U_{\text{liq}} + U_{\text{vap}}$$

$$U = (1-x)u_f + xu_g$$

$$U = U_f + x(U_g - U_f)$$

$$x = 0 \quad (\text{saturated liquid})$$

$$x = 1 \quad (\text{saturated vapour})$$

16 Oct 2024

Lecture 12

Enthalpy Enthalpy combines internal energy and pressure volume work. It is used while analyzing processes that occur at constant pressure. Applications involve boilers, turbines and compressors where heat transfer at constant pressure is common.

Derivation

Consider a control mass of gas undergoing equilibrium constant pressure work. The work is only done at boundary of the gas state as it expands or compresses. According to first law of thermodynamics

$$\begin{array}{ccccccc}
 U_2 - U_1 & = & Q_{12} & - & W_{12} & & \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 \text{Final} & & \text{Initial} & & \text{Heat transfer} & & \text{Work done} \\
 \text{internal} & & \text{internal} & & \text{from state 1} & & \text{from state 1} \\
 \text{energy} & & \text{energy} & & \text{to state 2} & & \text{to state 2}
 \end{array}$$

* Enthalpy is related to constant pressure

We assume a constant pressure process the work done is given by

$$W_{12} = P(V_2 - V_1)$$

Energy equation (First Law of thermodynamics) becomes

$$U_2 - U_1 = Q_{12} - P(V_2 - V_1)$$

$$Q_{12} = (U_2 - U_1) + P(V_2 - V_1)$$

Notice that expression for heat transfer involves a combination of internal energy and pressure volume work. We define a new property called Enthalpy as

→ enthalpy

$$H = U + PV$$

For constant pressure process, the heat transfer is

$$Q_{12} = H_2 - H_1$$

↓ final enthalpy
↓ initial enthalpy

Specific enthalpy

$$h = U + PV$$

Saturated and Superheated States

h_f : enthalpy of saturated liquid

h_g : enthalpy of saturated vapor

h_{fg} : enthalpy of saturated liquid vapor mixture

$$h = h_f + x \cdot h_{fg}$$

this is heat during the phase change of water from liquid to vapor (boiling)

Unit of specific enthalpy: kJ/kg
 ↳ per unit mass

Unit of Enthalpy: kJ (same as work)

Applications

Boilers and Steam turbines

Constant pressure, the heat add

$$Q_{\text{boiler}} = H_{\text{steam}} - H_{\text{water}}$$

Work done by the turbine

(no heat transfer only mechanical process takes place)

$$W_{\text{turbine}} = H_{\text{inlet}} - H_{\text{outlet}}$$

Heat Exchanges

$$\dot{Q} = \dot{m} (h_{out} - h_{in})$$

$\dot{m}$ is the mass flow rate

Internal Energy

$$U = H - PV$$

Enthalpy is a state property means it only depends on the state of the substances, not on the path taken to reach that state

Quiz Solution

QNO:1

a) Heat $500 - 300 = 200 \text{ kJ}$

b) work $200 - 100 = 100 \text{ kJ}$

c) $U = Q - W = 100 \text{ kJ}$

$\Delta U = 0$ as it is cyclic process

d)

e) Yes, it is a heat engine

QNO:2

a) $Q_{conv} = 3000 \text{ W}$

b) $Q_{rad} = \epsilon \sigma A T_s^4$
 $= () (400^4 - 300^4)$

f) $Q_{rad} = () (400^4 - 350^4)$

Lecture 13

Problem Analysis and Solution Technique

Steps

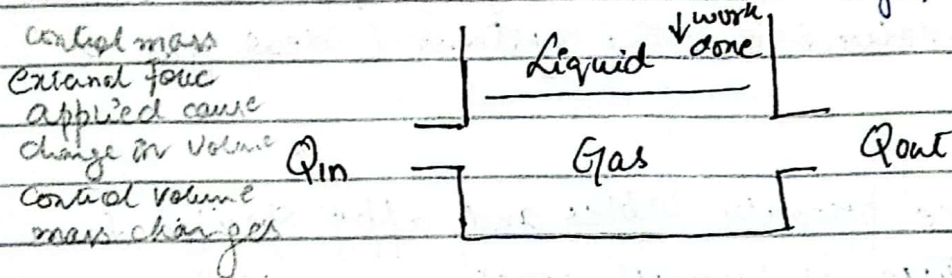
Step 1 Make a sketch of the system

Why? Helps in understanding how different aspects of system interacts

How? Indicate the direction of mass flow, heat transfer, work and external forces

Example

Piston cylinder system sketch the system to gas inside and allows show any flow of heat or work done



Step 2 Define a control mass or control volume

fixed amount of matter

specific space through which matter flows

Why? Defining the boundary of the system helps in applying the right law and equations

How? Place a closed surface around the system or subsystem you are analyzing. Identify any in flows or out flows of mass, energy and work

Step 3 Apply General conservation laws

Why? These are fundamentals to solve thermodynamics problem (conservation of energy, mass, first law of Thermodynamics)

How? Energy equation $\rightarrow$ ensure all inflows and outflows are mentioned

$$U_2 - U_1 = Q_{12} - W_{12}$$

Step 4 Write down any specific laws

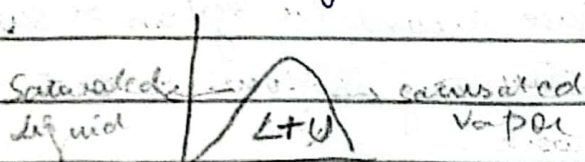
Why? These allow us to handle any specific details about the substances involved in the systems (ideal gas or steam etc)

How? Reference property tables and apply specific laws based on the nature of process (const volume, isothermal, adiabatic)

Steps Solve using diagrams or tables

Why? P-v, T-v diagrams help visualize the process

How? P-v, T-v diagrams visualization state determination



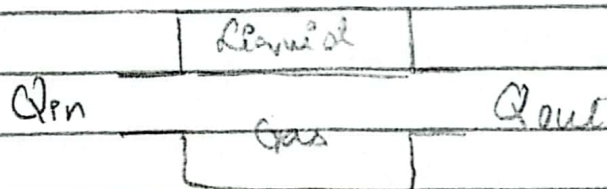
Step 1 Formulate and solve the equations

How? Combine all the equations and relationships then solve for unknown

Example $U_2 - U_1 = Q_{12} - P(V_2 - V_1)$ Adiabatic $\rightarrow Q = 0$

A vessel with a volume of 5 m^3 contains 0.05 m^3 of saturated liq. water and 4.95 m^3 of saturated vapour at 0.1 MPa . Heat is transferred to the system until it is filled with saturated vapour. Determine the heat transfer for this process.

Step 1



Initial Condition: Volume of saturated liquid = 0.05 m^3
Volume of saturated vapour = 4.95 m^3

Final State: Vessel is filled with saturated vapour at some higher pressure

Step 2 Control volume and control mass

Step 3

$$U_2 - U_1 = P_{12} - \cancel{(W_{12})} \rightarrow \text{as const volm}$$

$$U_2 - U_1 = P_{12}$$

Step 4

Saturated water pressure entry (Table B.12)
 $P_2 = 100 \text{ kJ}$

Find
 since we want
 to find mass

$$\begin{aligned} V_f &= 0.001043 \text{ m}^3/\text{kg} & , & \quad U_f = 417.36 \text{ kJ/kg} \\ V_g &= 1.6940 \text{ m}^3/\text{kg} & , & \quad U_g = 2506.1 \text{ kJ/kg} \end{aligned}$$

signs

Mass of Saturated liquid = 200 kg
 Mass of Saturated vapor = V_{vap} / V_g
 Mass of Saturated liquid = V_{liq} / V_f

$$\begin{aligned} m_{\text{vap}} &= \frac{V_{\text{vap}}}{V_g} \\ &= \frac{0.05}{0.001043} = 47.94 \text{ kg} \\ &= \frac{4.95}{1.6940} = 2.92 \text{ kg} \end{aligned}$$

Total mass = 50.86 kg

23 Oct 2024

Lecture 14

Conservation of Mass

"Mass cannot be created or destroyed"
control volume but not control mass (means mass enter and exit ho raha hoga)

For control volume, any change in mass inside the control mass volume must be accounted for any mass entering or leaving through the control surface

The equation of continuity is $\frac{dm_{cv}}{dt} = \sum \dot{m}_{in} - \sum \dot{m}_{out}$
 $\rightarrow$ rate of flow

If $\sum \dot{m}_{in} = \sum \dot{m}_{out}$ So $\frac{dm_{cv}}{dt} = 0$ this make control mass system (as mass is constant)

If $\frac{dm_{cv}}{dt} = 0$ Then mass remains constant

Mass flow rate Equation:

$$\begin{matrix} \text{(mass flow rate)} & \text{(density} \times \text{area)} \\ \dot{m} = \rho A V & \text{(velocity)} \\ \downarrow & \downarrow & \searrow & \searrow \\ \text{kg/s} & \text{kg/m}^3 & \text{m}^2 & \text{ms}^{-1} \end{matrix}$$

$\dot{m}$: mass flow rate

ρ : fluid density

A : cross sectional area

V : is the velocity of the fluid perpendicular to surface

$$\dot{m} = \frac{VA}{v}$$

$v \rightarrow$ specific volume

An ideal gas is flowing in a 0.2 m diameter pipe at a uniform velocity 0.1 m/s. The temperature is 25° pressure is 150 kPa

$$A = \pi r^2$$

$$A = \pi (0.1)^2$$

$$A = 0.0314 \text{ m}^2$$

As ideal gas

$$PV = nRT$$

(volume has same mass mR)

as specific volume ideal gas

$$Pv = RT$$

$$v = \frac{RT}{P}$$

$$R = 0.287 \text{ kJ/kgK (air)}$$

$$T = 25 + 273 = 298 \text{ K}$$

$$= \frac{0.287 \times 298}{150}$$

$$= 0.57017$$

$$v = 0.57017 \text{ m}^3/\text{kg}$$

$$\dot{m} = \frac{VA}{v}$$

$$= \frac{(0.1)(0.0314)}{0.57017}$$

$$\dot{m} = 0.0055 \text{ kg/s}$$

Ch #4

Topic 4-1

Energy Equation for a control volume in thermodynamics

For control mass the energy equation is written as

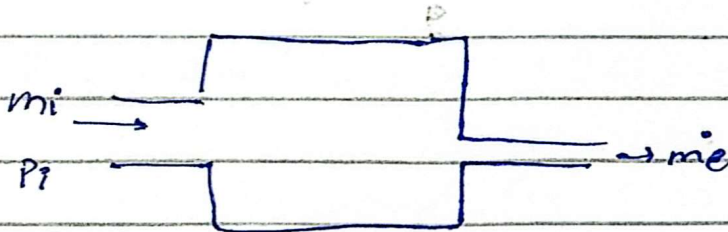
$$E_2 - E_1 = Q_{12} - W_{12}$$

Instantaneous form

$$\frac{dE_{cm}}{dt} = \dot{Q} - \dot{W}$$

The energy equation for control volume

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum \dot{m}_i (e_i + P_i v_i) - \sum \dot{m}_e (e_e + P_e v_e)$$



where

$$e = U + \frac{V^2}{2} + gZ \quad (\text{mass is unit mass})$$

Internal energy

KE

PE

Pv is work flow rate

Subscripts:

'i' → incoming

'e' → outgoing

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum \dot{m}_i \left(u_i + \frac{v_i^2}{2} + g z_i + P_i v_i \right) - \sum \dot{m}_e \left(u_e + \frac{v_e^2}{2} + g z_e + P_e v_e \right)$$

the total energy per unit mass is

$$u + P v + \frac{v^2}{2} + g z$$

Specific enthalpy $(h) + \underbrace{\left(\frac{v^2}{2} + g z \right)}_{\text{Specific kinetic energy}}$

The energy equation for control volume

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum \dot{m}_i \left(h_i + \frac{v_i^2}{2} + g z_i \right) - \sum \dot{m}_e \left(h_e + \frac{v_e^2}{2} + g z_e \right)$$

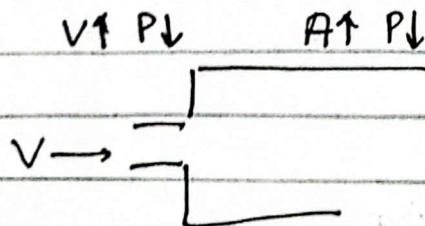
Applications

Turbines $\rightarrow$ mechanical work ($\dot{W}_{cv}$)

Compressions $\rightarrow$ Pressure & volume

Heat exchange $\rightarrow$ heat ($\dot{Q}_{cv}$)

Nozzles $\rightarrow$ (pressure, velocity, area)



Steady State Process:

where the properties of the system do not change with time

- $\dot{m}_i$ and $\dot{m}_e$ are constant
- properties of fluid entering or existing are constant
- Energy within control volume does not accumulate over time

The energy equation for control volume

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \sum \dot{m}_i \left(h_i + \frac{v_i^2}{2} + g z_i \right) - \sum \dot{m}_e \left(h_e + \frac{v_e^2}{2} + g z_e \right)$$

Turbine:

$$\dot{W}_{cv} = \dot{Q}_{cv} + \sum \dot{m}_i \left(h_i + \frac{v_i^2}{2} + g z_i \right) - \sum \dot{m}_e \left(h_e + \frac{v_e^2}{2} + g z_e \right)$$

→ multiple input and output
as $\sum \dot{m}_i = \sum \dot{m}_e = \dot{m}$ (as constant)

$$\dot{W}_{cv} = \dot{Q}_{cv} + \sum \dot{m} \left(h_i + \frac{v_i^2}{2} + g z_i \right) - \sum \dot{m} \left(h_e + \frac{v_e^2}{2} + g z_e \right)$$

$$\dot{W}_{cv} = \dot{Q}_{cv} + \sum \dot{m} \left(h_i + \frac{v_i^2}{2} + g z_i - h_e - \frac{v_e^2}{2} - g z_e \right)$$

$$\dot{W}_{cv} = \dot{Q}_{cv} + \dot{m} \left((h_i - h_e) + \frac{1}{2} (v_i^2 - v_e^2) + g (z_i - z_e) \right)$$

power power wall wall
→ $\dot{Q}/s = \dot{W}_{wall}$

No work is done in nozzle

Lecture 15

Steady State process

1. Control volume is stationary boundary of the control volume does not move
2. mass constant

$$\frac{dm_{cv}}{dt} = 0 \quad \text{and} \quad \frac{dE_{cv}}{dt} = 0$$

$$\sum \dot{m}_i = \sum \dot{m}_e$$

$$Q_{cv} + \sum \dot{m}_i \left(h_i + \frac{v_i^2}{2} + g z_i \right) = \sum \dot{m}_e \left(h_e + \frac{v_e^2}{2} + g z_e \right) + \dot{W}_{cv}$$

Energy Equation: $\dot{Q}_{cv} + \dot{m} \left(h_i + \frac{v_i^2}{2} + g z_i \right) = \dot{m} \left(h_e + \frac{v_e^2}{2} + g z_e \right) + \dot{W}_{cv}$

$$q = \frac{Q_{cv}}{\dot{m}} \quad \text{and} \quad w = \frac{W_{cv}}{\dot{m}}$$

Specific heat and work = kJ/kg

$$\dot{w} = \frac{W}{t} = \frac{J}{s} = W$$

$$q = \frac{Q_{cv}}{\dot{m}} \quad q = \frac{Q/t}{m/t} = \frac{Q}{m}$$

$$\dot{Q} = \frac{Q}{t} \quad \dot{m} = \frac{m}{t}$$

6 NOV

Lecture 16:

Heat Engine

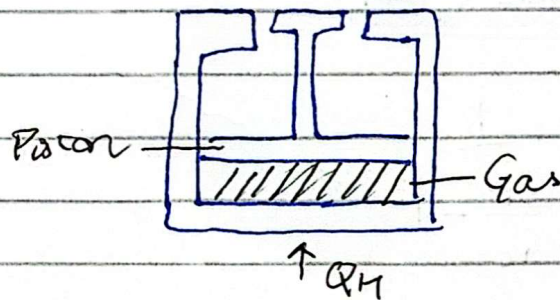
A device that operates in a thermodynamics cycle transferring heat from high temperature reservoir to a low temperature reservoir and producing a net positive work

- Steam, gas or any other substance undergoes a cyclic process of heating and cooling

Topic 5.1 Heat Engines & Refrigerators

Steam power plant / engine

- Heat Q_H is absorbed from high temp source
- Steam flows through the turbine - performing work
- Steam is condensed, releasing Q_L heat to a low temp reservoir



Thermal Efficiency

It is a measure of how well the heat engine converts heat input into useful work

$$\eta_{\text{thermal}} = \frac{W}{Q_H} = 1 - \frac{Q_L}{Q_H}$$

Q_H heat absorbed
 Q_L heat released

$$W = Q_H - Q_L$$

Efficiency of real engine

Large power plant 35-50%
Gasoline engine 30-35%
Diesel engine 30-40%
Smaller engine 20%

$$\eta_{\text{thermal}} = \frac{W}{Q_H} = 50\%$$

$$W = 50\% \times Q_H = 0.5 Q_H$$

$$Q_L = Q_H - W \\ = Q_H - 50\% Q_H$$

$$Q_L = Q_H (1 - 50\%) \\ = Q_H (1 - 0.5)$$

$$Q_L = 0.5 Q_H$$

Refrigerator and Heat Pumps

Refrigerator is a device that uses work to transfer heat from a lower temperature to a higher temperature one. It operates on a cycle similar to a heat pump with a focus on cooling.

$$\eta_{\text{thermal}} = \frac{W}{Q_H} = 20\%$$

$$W = 20\% \text{ of } Q_H = 0.2 Q_H$$

$$Q_L = Q_H - W \\ = Q_H - 0.2 Q_H$$

$$Q_L = 0.8 Q_H \quad \text{worst Heat engine}$$

Coefficient of performance (COP)

$$B = \frac{Q_L}{W} = \frac{Q_L}{Q_H - Q_L} = \frac{1}{\frac{Q_H}{Q_L} - 1}$$

Basic Cycle

- A typical refrigerator uses a vapour temp cycle
- Refrigerant R-134a or ammonia absorbs Q_L in the evaporator at low pressure and temperature
- Refrigerant releases Q_H in the condenser

Heat pump When the objective is heat a space rather than cool it is called heat pump

$$COP = \beta' = \frac{Q_H}{W} = \frac{Q_H}{Q_H - Q_L}$$

for heat pump

Relation b/w η & β

$$\eta_{\text{thermal}} = \frac{W}{Q_H} \quad \beta = \frac{Q_L}{W} \Rightarrow W = \frac{Q_L}{\beta}$$

$$\eta = \frac{Q_L}{Q_H} \quad \beta = \frac{Q_L}{Q_H \times \eta}$$

Relation b/w β and β'

$$\beta = \frac{Q_H}{W} \quad \beta' = \frac{Q_L}{W} \Rightarrow W = \frac{Q_L}{\beta}$$

$$\beta' = \frac{Q_H}{\frac{Q_L}{\beta}} \quad \beta' = \frac{Q_H \beta}{Q_L} \quad \beta = \frac{Q_L \beta'}{Q_H}$$

- Reservoir from which heat is transferred it is called source
- Reservoir to which heat is transferred is called sink

Lecture 17

Second Law of Thermodynamics

Governs energy transfer and conversion in thermodynamics system
Introduces concept of irreversibility and efficiency

The Law has two classical statements — Kelvin Planck
Clausius Statement

Kelvin Planck Statement

It is impossible to construct a device that operates on a cycle and produces no effect other than the raising of a weight (work done) and the exchange of heat with a single thermal reservoir (all at T temperature)

Key implications

→ Heat engine cannot be 100% efficient i.e. $\eta \neq 100\%$.

$$W = Q_H - Q_L$$

→ $\eta = \frac{W}{Q_H} \neq 100\%$

Q_L is always smaller than Q_H
 $Q_L < Q_H$

→ Heat must flow from a high temp to a low temp
reversible

heat rejected
if Q_H is heat input
so Q_L = heat rejection

if $Q_L = 0$
that means
all heat W
from Q_H

Clausius Statement It is impossible to construct a device that operates in a cycle and produces no effect other than the transfer of heat from a cooler body to a warmer body.

Refrigerator / Heat pump Heat cannot flow from cold reservoir to a hot reservoir without work done

$$COP_{ref} = \frac{Q_L}{W} = \frac{Q_L}{Q_H - Q_L} \rightarrow \text{heat transfer from cooler to hotter body}$$

$$COP_{heat\ pump} = \frac{Q_H}{W} = \frac{Q_H}{Q_H - Q_L}$$

Violation of Clausius Statement implies violation of Kelvin Planck Statement

The reverse is also true

Perpetual Motion Machines

First kind Creates energy from nothing violating the first law of thermodynamics

Second kind Converts heat flow from a single reservoir entirely into work, violating the second law $| Q_H = W |$

Third kind operates indefinitely without friction of work output theoretically impossible

Thermal Reservoir:

↳ High temperature

↳ Low temperature

Heat engine

If $T_H - T_L \uparrow$

$\eta \uparrow$

As $W = Q_H - Q_L$

$\uparrow \eta = \frac{W}{Q_H} \uparrow$

Reversible Process A reversible process is an idealized process that, when reversed, leaves both the system and surroundings completely unchanged.

→ In reality, all processes are irreversible.

→ Irreversible processes deviate from equilibrium.

Factors of Irreversibility

1. Friction

2. Unrestrained Expansion.

Restricted is reversible process

3. Heat transfer through a finite temp difference

4. Mixing of different substance

Internal irreversibility (within a system) (one is involve either system or surroundings)

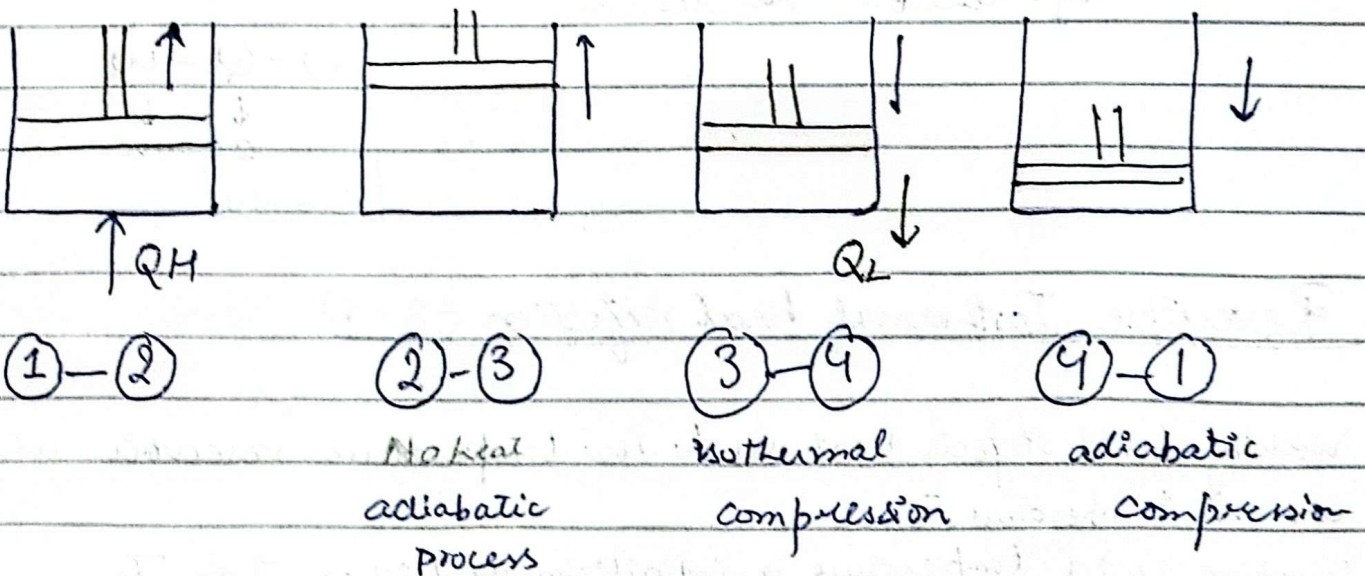
External irreversibility (external condition) (Surrounding & system both involve)

Lecture 18

Carnot Cycle

Carnot cycle consist of 4 reversible process

1. Reversible isothermal heat addition at a high temperature
2. Reversible adiabatic expansion from the high temperature to the low temperature
3. Reversible isothermal heat rejection at a low temperature
4. Reversible adiabatic compression back to the high temperature



1. Reversible isothermal heat addition at a high temperature

- Absorb Q_H from high temp reservoir
- Heat transfer takes place
- Working fluid expands, doing work on the surrounding
- May involve a phase change
- $Q_H = T_H (\Delta S)$ → Change in entropy
- Q and W are positive

$$U = Q - W$$

$\downarrow$ $\downarrow$
 +ve +ve
 $\downarrow$
 Region
 to
 piston
 work is
 done on
 the surrounding

2. Reversible Adiabatic Expansion (2-3)

- Working fluid expands without exchanging heat with the surrounding
- Adiabatic process $Q = 0$
- Temperature of the fluid decreases from T_H to T_L
- Work is done by the system on the surroundings

$$T_H V_2^{\gamma-1} = T_L V_3^{\gamma-1}$$

$\downarrow$ $\downarrow$
 High reservoir Low reservoir

for ideal gas

$$\gamma = \frac{C_p}{C_v}$$

$$U = Q - W$$

$\downarrow$ $\downarrow$
 0 +ve

$$\boxed{U = -W}$$

3. Reversible Isothermal heat rejection (3-4)

Piston is coming back to its position after heat expansion, now the no gas absorption happens

- Working fluid rejects heat Q_L to low temperature reservoir at constant temperature T_L
- Working fluid temperature is infinitesimally higher than T_L
- The process is isothermal
- Heat is removed from working fluid causing partial condensation or compression

Expansion Stop
Pressure equal to piston pressure

$$Q_L = T_L \Delta S$$

$$U = Q - W$$

$\downarrow$
 +ve

4. Reversible Adiabatic Compression (4-1)

— The working fluid is compressed adiabatically raising temperature back to T_H

— Conditions:

Adiabatic process $Q=0$

Reversible work done on the system by surrounding

— Effect on system:

System temperature raise from T_L to T_H

$$T_L V_4^{\gamma-1} = T_H V_1^{\gamma-1}$$

$$U = Q - W$$

$\quad \quad \quad 0 \quad \quad \text{+ve}$

$$\boxed{U = +W}$$

Efficiency of Carnot cycle:

$$\eta = 1 - \frac{Q_L}{Q_H} = 1 - \frac{T_L}{T_H}$$

Key features

— Reversible

— Depends only on the temperature

— Entropy $\Delta S_H = \frac{Q_H}{T_H}$, $\Delta S_L = \frac{Q_L}{T_L}$

Applications

- Real world limitations
- No real engine can operate on the Carnot cycle due to irreversibilities
- Carnot cycle is just a benchmark for real engine.

Two propositions regarding the efficiency of a Carnot cycle

1/ It is impossible to construct an engine that operates between two given reservoirs and is more efficient than a reversible engine operating between the same two reservoirs.

$$\eta_{\text{any}} \leq \eta_{\text{reservoir}}$$

η_{any} → efficiency of any engine operating between two reservoirs

2/ All engines that operate on the reversible Carnot cycle between two given constant-temperature reservoirs have the same efficiency.

$$\eta_{\text{rev1}} = \eta_{\text{rev2}}$$

Lecture 20

State 1-2

State 2-3 pressure decrease v

State 3-4 isothermal compression

State 4-1 pressure $\uparrow$ volume $\downarrow$ adiabatic compression

$$\frac{Q_H}{Q_L} = \frac{RT_H \ln \frac{v_2}{v_1}}{RT_L \ln \frac{v_3}{v_4}} = \frac{T_H}{T_L}$$

Topic 5.9

$$(\text{Refrigerator}) B = \frac{Q_L}{Q_H - Q_L} = \frac{T_L}{T_H - T_L}$$

$$(\text{Heat pump}) B' = \frac{Q_H}{Q_H - Q_L} = \frac{T_H}{T_H - T_L}$$

for Carnot cycle
↳ Reversible ProcessCondition
Initial & final sameState not a full cycle
four steps

for real (irreversibilities involved)

Example

$$W = Q_H - Q_L$$

$$\dot{W} = P = \frac{Q_H}{t} - \frac{Q_L}{t}$$

Carnot → ideal heat engine

$$\eta = \frac{W}{Q_H} = \left[\frac{\dot{W}}{\dot{Q}_H} \right] \rightarrow \text{power form}$$

20 NOV 2024

Lecture 21

Chapter 6 Entropy

$$\oint \frac{Q}{T} \leq 0 \quad (\text{Clausius statement})$$

This inequality applies to all possible thermodynamics cycles - whether reversible or irreversible and is used to define entropy

Entropy is define as $ds = \frac{\delta Q}{T} \Big|_{\text{rev}}$

Entropy change $S_2 - S_1 = \int_1^2 \frac{\delta Q}{T} \Big|_{\text{rev}}$

Reversible Heat Engine cycle

Net Heat transfer $\oint \delta Q = Q_H - Q_L > 0$

Net entropy change $\oint \frac{\delta Q}{T} = \frac{Q_H}{T_H} - \frac{Q_L}{T_L} = 0$

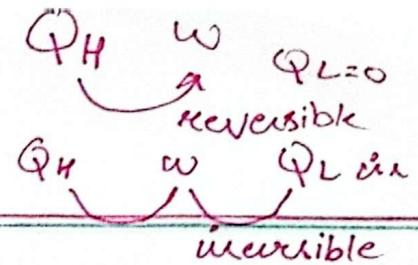
For all reversible cycles

$$\boxed{S_2 - S_1 = \oint \frac{\delta Q}{T} = 0}$$

Irreversible Heat Engine cycle

Work: $W_{\text{irr}} < W_{\text{rev}}$

Heat transfer $Q_L^{irr} > Q_L^{rev}$



Net Entropy change $\oint \frac{Q}{T} = \frac{Q_H}{T_H} - \frac{Q_L^{irr}}{T_L} < 0$

For all irreversible cycle $S_2 - S_1 = \oint \frac{\delta Q}{T} < 0$

High entropy mean energy is scattered and things are more disorder and chaotic

Low entropy mean energy is more concentrated and organize

Entropy Randomness & disorder on the system

when heat flows in the system

$Q \rightarrow$ [System] $\rightarrow$ when heat flow out of the system (S-ve)
 when heat flows into the system entropy increases (S+ve)
 $\rightarrow$ entropy increase

Natural phenomena entropy (decrease) as it flow from high to low
 $\rightarrow$ as it attain stability

Refrigerator

Heat transfer $\oint \delta Q = -Q_H + Q_L < 0$

Net entropy change $\oint \frac{\delta Q}{T} = -\frac{Q_H}{T} + \frac{Q_L}{T}$

Irreversible Refrigeration Cycle

work input $W_{irr} > W_{rev}$

$$Q_H^{in} > Q_H^{out}$$

Net entropy change

$$\oint \frac{\delta Q}{T} = -\frac{Q_H^{in}}{T} + \frac{Q_L}{T} < 0$$

$$\oint \frac{\delta Q}{T} = 0 \quad (\text{reversible cycle})$$

$$\oint \frac{\delta Q}{T} < 0 \quad (\text{irreversible cycle})$$

Absolute zero $\rightarrow -273^\circ\text{C} \rightarrow 0 \text{ Kelvin}$

$\hookrightarrow$ Entropy zero

Specific Entropy ($\text{kJ/kg}\cdot\text{K}$)

Entropy
 $\downarrow$
 kJ/K

Saturated Region

$$S = (1-x)S_f + xS_g$$

$$S = S_f + xS_{fg}$$